CCT382 - Milestone 2: Hero Mechanic

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Audio Citations:

Cat walking: Sound Effect by freesound_community from Pixabay

Cat growl and hiss: Sound Effect by freesound_community from Pixabay

Swat (miss): Sound Effect by freesound_community from Pixabay

Swat (hit): Sound Effect by Dilip from Pixabay

Mouse hurt: Sound Effect by freesound_community from Pixabay

Mouse block: Sound Effect by freesound_community from Pixabay

Mouse parry: Sound Effect by Ryan Lewis from Pixabay

Block cooldown icon: designed by Andie

Distribution of Work:

- Michaela
    - Cat movement scripts/animations
    - Mouse movement scripts/animations
    - Attack/damage visual feedback
    - Sound effects
- Andie
    - Health scripts
    - Health bar visual feedback
    - Health bar and block cooldown UI
    - Attack wind-up visual effect

For Milestone 2, we implemented more movement scripts, UI design, and sound and visual feedback. We created an AI movement script for the cat that includes a resting state (resting on the cat couch cushion), a chasing state (when the mouse is in view, it chases and attacks the cat), and a returning state (when the mouse is out of view, it returns to rest position). Using ShaderGraphs, we implemented visual feedback for hits/damage: the cat flashes white when it attacks, and the mouse flashes red when it receives damage. The cat's paw also has a trail

effect when it is about to attack, and the mouse has a particle system effect when it sprints. The mouse has a blocking ability (hero mechanic) that allows the mouse to block damage, and when blocked at the right time and distance from the cat, it stuns the cat to sleep and allows the mouse to escape. This ability is triggered by a right-click input from the player. This ability has a small cooldown that is visually displayed by a block icon in the bottom right corner that has a radial grey-out effect when the ability is on cooldown. We have a health script that uses PlayerHealth as a ScriptableObject, allowing the mouse to take damage from the cat's attack, take partial damage if the mouse blocks imperfectly, and receive no damage if the mouse blocks perfectly. To communicate this visually to the player, the health bar shakes violently and flashes red when the mouse takes damage, and it shakes a little and flashes yellow when it takes partial damage from blocking. The game's environment uses NavMesh. We improved the camera slightly using Cinemachine and some scripts so the camera now rotates in the direction the mouse is moving (e.g. if the camera is facing towards the mouse and the player inputs the command to move forward, the camera rotates so that the camera is now looking in the same direction that the mouse is moving forward). And of course, we kept the same core engine systems as the first milestone. Additionally, another element we expanded on for this milestone is the use of sound.

In the week 4 reading, "Sound for Fantasy and Freedom" in *In Game Sound Technology and Player Interaction: Concepts and Developments*, it discusses how audio is a very useful feedback tool. The human mind has a higher threshold for auditory sensory input than it does for visual sensory input. As a result, sound can relay a lot of information to the user without overwhelming them. Thus, in addition to some visual feedback of damage, we actually implemented a lot of sound feedback. There are different sound effects when the mouse gets hurt, when it blocks, and when it blocks perfectly. There are also different sounds for when the cat's attack hits or if it misses, as well as a growl sound that indicates the cat is winding up to attack. These auditory cues subconsciously give the player this information. Supplemented by some visual non-language cues, this gives players a lot of useful information to play the game while not being too overwhelmed by a lot of text.

One of the most difficult technical hurdles for this milestone was timing the audio and animations to the script. For example, the cat had 0.6 seconds of windup. It was diffcult to align the animation and sound to fit that windup time. This was solved by cutting down the animation time and sound time through the script and adding extra variables that can be changed. A

moment of compromise due to the limitations of Unity's underlying architecture in this milestone was the feel of the game. Because of the NavMesh and details with the CatAI, it was impossible to get the feel of the cat to be the exact same as a real cat. We had to deal with the outcome of what it looks like as is. It makes it feel more cartoony and slightly AI, but it does work and gets the job done. Overall, we are very happy with how this milestone turned out.